

Dipayan Sarkar

Computational Biology | Generative AI for Protein Design

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Summary

PhD researcher in bioinformatics and computational biology developing deep learning methods for protein sequence modeling. First author on four papers spanning attention-based protein-protein interaction prediction, uncertainty calibration, and generative binder design. Pretrained a domain-adapted protein language model family on 1.1B amino-acid tokens, and build web platforms that make trained models usable for inference and network visualisation.

Skills

Languages	Python, R.
ML / DL	PyTorch, Hugging Face Transformers, attention architectures, protein language models, domain-adaptive pretraining, transfer learning, autoregressive generative models, uncertainty calibration, mixture-of-experts, scikit-learn, NumPy, Pandas
Scale	Multi-GPU & HPC training, Slurm, mixed precision, gradient checkpointing, CUDA, NVIDIA L4/A100, Cloud TPU (v4/v5e/v6e), Weights & Biases
Domain	Protein-protein interaction & biomolecular networks, de novo binder design, sequence analysis, MMseqs2, UniProt
Engineering	Docker, FastAPI, Flask, Nginx, Linux server deployment.

Experience

PhD Research Scholar, Department of Bioinformatics, University of North Bengal Siliguri, India | 2022-Present

- Designed **AttnSeq-PPI**, a transfer-learning hybrid attention architecture for sequence-based protein-protein interaction prediction; published in *BBA - Proteins and Proteomics* and released as a public web server.
- Built **ARACoFusion**, an uncertainty-aware calibrated ensemble for PPI network prediction in *Arabidopsis thaliana*, optimising for reliable confidence estimates rather than raw accuracy alone.
- Developed **MoE-Bind**, a sparse mixture-of-experts autoregressive generative model for de novo protein binder design.
- Pretrained **PlantPLM**, a family of domain-adapted ESM-2 protein language models (8M/35M/150M), on 1.11B amino-acid tokens from 4.37M MMseqs2-deduplicated Viridiplantae sequences, reaching 8.98 validation perplexity. Trained end-to-end on a single 12 GB consumer GPU using mixed precision, gradient checkpointing, and gradient accumulation.
- Shipped and maintain three model-backed web platforms (Docker + FastAPI/Flask + Nginx) serving inference and interaction-network visualisation to external users.

Open-Source Software & Model Releases

PlantPLM	Domain-adapted ESM-2 protein language models (8M / 35M / 150M) continued-pretrained on plant proteomes; released with weights, training dataset, and interactive demo. huggingface.co/dipayanskr
Deep-Interact Studio	Web platform to build, train, and compare deep learning models for protein-protein, drug-target, RNA-protein, and protein-DNA interaction prediction, with integrated interpretability. deepinteract.compbiosysnbu.in
AttnSeq-PPI	Deep learning protein-protein interaction prediction server. attnseqppi.compbiosysnbu.in
EGRNi	R package on CRAN (v0.1.6) for ensemble gene regulatory network inference; 9,600+ downloads. cran.r-project.org/package=EGRNi

Selected Publications

Sarkar, D., & Sarkar, C. (2025). AttnSeq-PPI: Enhancing protein-protein interaction network prediction using transfer learning-driven hybrid attention. *BBA - Proteins and Proteomics*, 1874(1), 141102.

Sarkar, D., & Sarkar, C. (2026). MoE-Bind: Guiding de novo protein binder generation with sparse experts. *bioRxiv*.

Sarkar, D., & Sarkar, C. (2026). ARACoFusion: Uncertainty-aware calibrated deep learning for PPI network prediction in *Arabidopsis thaliana*. *bioRxiv*.

Sarkar, D., Bardhan, K., & Sarkar, C. (2026). Deep-Interact Studio: An interactive deep learning model building platform for biomolecular interaction prediction. *bioRxiv*.

Education

Ph.D. in Bioinformatics, University of North Bengal India | Expected submission 2027

M.Sc. in Botany (Specialisation: Biochemistry), University of North Bengal India | 2021

Awards & Compute Grants

CSIR-UGC NET with JRF, Life Sciences (2021): All India Rank 216. • **GATE** Life Sciences (XL), qualified (2022).

IndiaAI Compute Initiative (2025): NVIDIA L4 GPU cloud compute for PPI network prediction (Sep 2025 to Aug 2026).

Google TPU Research Cloud (TRC) (2025): Cloud TPU access (v4, v5e, v6e) for machine learning research.